

Lesson 17 • 40 marks • Suggested time: 45 minutes

Instructions

Work independently with the experience and notes closed. Answer every question. Show the steps in searches and sorts. Use the marks to judge how much detail to include. Questions are original practice, not an official OCR paper.

1 Computational thinking [6]

A school is designing an equipment-loan system.

- a) Define abstraction. [2]
- b) Give one detail the system should retain and one it can ignore. [2]
- c) Explain how decomposition helps develop the system. [2]

2 Structure diagram [4]

Draw a structure diagram for the equipment-loan system. It must include Members, Loans and Returns. Break Loans into two appropriate smaller tasks. [4]

3 Searching [6]

The list is [2, 4, 6, 8, 10, 12, 14]. Target = 12.

- For a left-to-right linear search, list the values checked and give the comparison count. [2]
- For binary search, use zero-based indices and round midpoints down. State the initial midpoint value, the new index range and the next checked value. [3]
- State the data requirement for binary search. [1]

4 Sorting [10]

- a) Show the first full left-to-right bubble-sort pass on [6, 2, 5, 1]. Show each swap, then identify the value in its final position. [4]
b) Merge sorted lists [2, 8] and [3, 7]. State the first two comparisons and the final list. [3]
c) Show the whole list after each insertion when insertion-sorting [5, 2, 4, 1]. [3]

5 Create a flowchart [5]

Draw a flowchart that inputs a pupil age. If age is at least 12, output Allowed. Otherwise output Too young. Include a start, end, appropriate symbols and labelled branches. [5]

6 Debug an algorithm [3]

A programmer wants the average of a and b. The line is $\text{average} = a + b / 2$.

- a) State whether this is a syntax or logic error. [1]
b) Correct the expression and explain why your correction works. [2]

7 Complete a trace table [6]

Trace this OCR-style pseudocode.

```
total = 0
for i = 1 to 4
    total = total + i
next i
print(total)
```

Record the initial total, the total after each iteration and the final output. [6]

Stage / i	total	Output
Initial		
1		
2		
3		
4		
After loop		

Total: _____ / 40

Reflection after feedback

Complete this page when your marked assessment is returned.

Question	Available	My mark	Next lesson to revisit
Computational thinking	6		1–3
Structure diagram	4		3
Searching	6		7–8
Sorting	10		9–10
Create a flowchart	5		11
Debug an algorithm	3		15
Complete a trace table	6		16

My next step

Choose one missed question. Write a corrected answer and explain the mistake you will avoid next time.

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